



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science

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Project Report On

Project Name Fuel Management System

Supervised By

Dr. Md. Iftekharul Mobin

Submitted By:

Name			ID		
1. S. M. Tanvir Hassan			24-58617-2		
2. Kazi Sajidul Islam			24-58645-2		
Obtained Marks for CO2 and CO3 (Description given in the following page)					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)	Total =		Evaluation Criteria (CO3)		Total =
Requirement fulfillment			Organization of the application		
Validation			Representation and Integration of Database		
Verification			Graphical User Interface		

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1.Title

FuelMS : Fuel Management System

2. INTRODUCTION

Project Overview

FuelMS (Fuel Management System) is a comprehensive Windows desktop application designed to automate and streamline fuel station operations. The system provides a role-based management platform that handles fuel inventory, customer orders, payments, and administrative tasks through an intuitive graphical user interface.

Background

Traditional fuel station management often relies on manual processes or disconnected systems, leading to inefficiencies in inventory tracking, order processing, and customer management. FuelMS addresses these challenges by providing a centralized digital solution that integrates all aspects of fuel station operations.

Objectives

- Automate fuel inventory management with real-time stock tracking
- Provide role-based access control (Super Admin, Admin, Customer)
- Enable online fuel ordering for registered customers
- Track monthly fuel limits for customers

- Generate comprehensive reports and activity logs
- Facilitate digital payments and order approvals

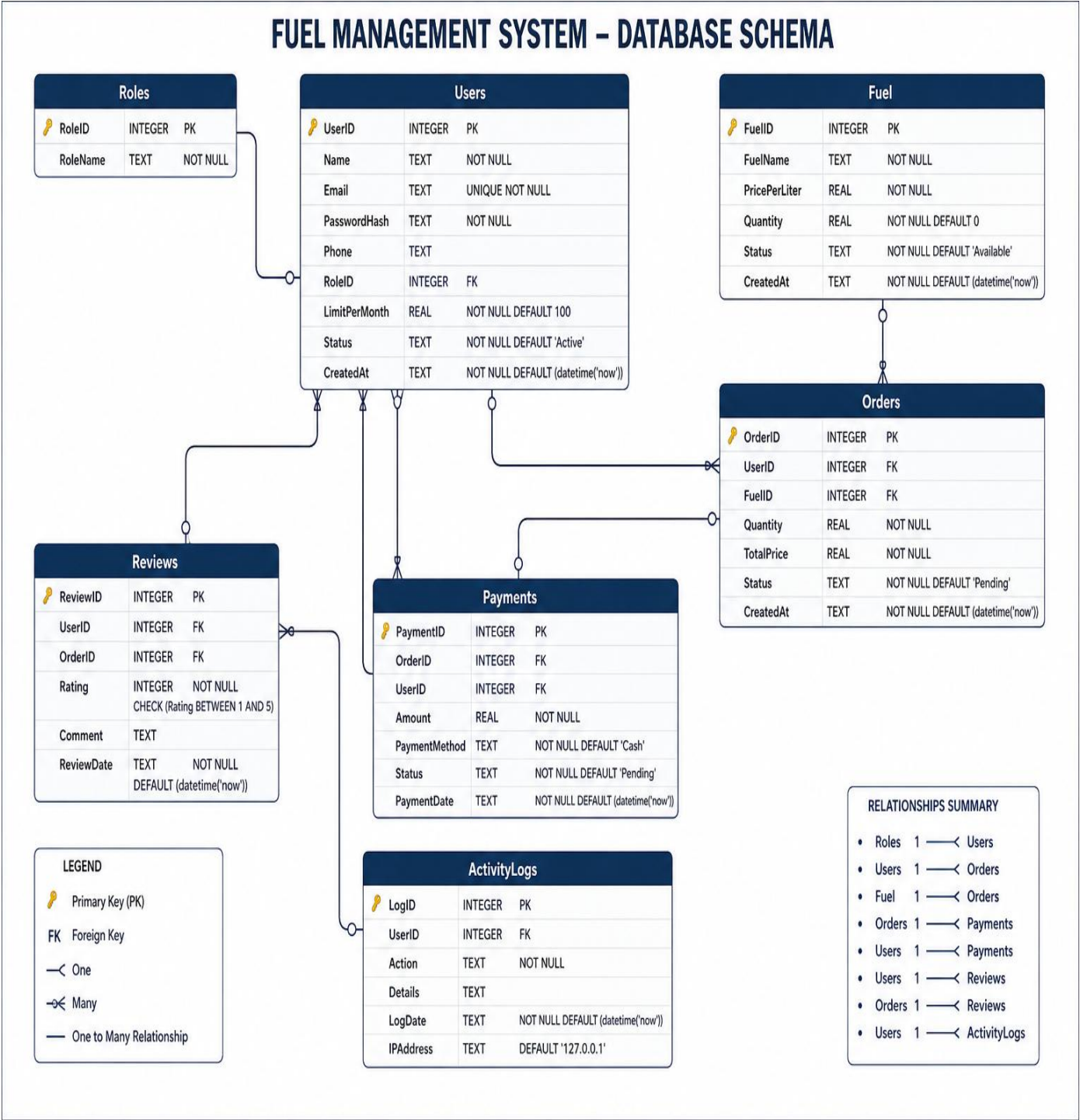
3. Case Study: Fuel Management System

In a fuel management system, users can view available fuel types, place orders based on quantity, and make payments using cash or online methods. Each user is identified by details such as user ID, name, email, phone number, and address. Each fuel type has a unique ID and includes information like name, price per liter, and available stock. The system automatically updates stock after each transaction. Orders are recorded with details such as order ID, user ID, fuel type, quantity, total price, and status (pending, approved, or rejected). Administrators manage the system by controlling fuel prices, updating stock, approving orders, and generating reports. Users can also provide reviews and track their order history, ensuring a smooth and efficient fuel management process.

4. FUNCTIONALITIES:

1. After login, a user can view available fuel types with price and quantity.
2. User can place an order by selecting fuel type and quantity.
3. The system calculates total price based on quantity and price per liter.
4. User can make payment for the order.
5. User can view order status (e.g., Pending, Completed).
6. User can see their order and payment history.
7. User can give rating and review for an order.
8. Admin can add and manage fuel types (price, quantity, status).
9. Admin can monitor and update order status.
10. Admin can track and manage payments.
11. Admin can view and manage users.
12. Super Admin can manage all users and admins (full control).
13. Super Admin can monitor overall system activities and data.
14. System records activity logs of user actions.
15. System automatically stores date and time for orders, payments, and logs.

5. DATABASE SCHEMA:



6. DATABASE QUERIES

Table Creation Queries

Roles Table:

```
CREATE TABLE IF NOT EXISTS Roles (  
    RoleID INTEGER PRIMARY KEY,  
    RoleName TEXT NOT NULL
```

Users Table:

```
CREATE TABLE IF NOT EXISTS Users (  
    UserID INTEGER PRIMARY KEY AUTOINCREMENT,  
    Name TEXT NOT NULL,  
    Email TEXT UNIQUE NOT NULL,  
    PasswordHash TEXT NOT NULL,  
    Phone TEXT,  
    RoleID INTEGER NOT NULL DEFAULT 3,  
    LimitPerMonth REAL NOT NULL DEFAULT 100,  
    Status TEXT NOT NULL DEFAULT 'Active',  
    CreatedAt TEXT NOT NULL DEFAULT (datetime('now')),  
    FOREIGN KEY (RoleID) REFERENCES Roles(RoleID)
```

Fuel Table:

```
CREATE TABLE IF NOT EXISTS Fuel (  
    FuelID INTEGER PRIMARY KEY AUTOINCREMENT,  
    FuelName TEXT NOT NULL,  
    PricePerLiter REAL NOT NULL,
```

Quantity REAL NOT NULL DEFAULT 0,
Status TEXT NOT NULL DEFAULT 'Available',
CreatedAt TEXT NOT NULL DEFAULT (datetime('now'))
);

Orders Table:

```
CREATE TABLE IF NOT EXISTS Orders (  
    OrderID INTEGER PRIMARY KEY AUTOINCREMENT,  
    UserID INTEGER NOT NULL,  
    FuelID INTEGER NOT NULL,  
    Quantity REAL NOT NULL,  
    TotalPrice REAL NOT NULL,  
    Status TEXT NOT NULL DEFAULT 'Pending',  
    CreatedAt TEXT NOT NULL DEFAULT (datetime('now')),  
    FOREIGN KEY (UserID) REFERENCES Users(UserID),  
    FOREIGN KEY (FuelID) REFERENCES Fuel(FuelID)  
);
```

Payments Table

```
CREATE TABLE IF NOT EXISTS Payments (  
    PaymentID INTEGER PRIMARY KEY AUTOINCREMENT,  
    OrderID INTEGER NOT NULL,  
    UserID INTEGER NOT NULL,  
    Amount REAL NOT NULL,  
    PaymentMethod TEXT NOT NULL DEFAULT 'Cash',
```



```
Status TEXT NOT NULL DEFAULT 'Pending',  
PaymentDate TEXT NOT NULL DEFAULT (datetime('now')),  
FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),  
FOREIGN KEY (UserID) REFERENCES Users(UserID)  
);
```

Reviews Table:

```
CREATE TABLE IF NOT EXISTS Reviews (  
ReviewID INTEGER PRIMARY KEY AUTOINCREMENT,  
UserID INTEGER NOT NULL,  
OrderID INTEGER NOT NULL,  
Rating INTEGER NOT NULL CHECK(Rating BETWEEN 1 AND 5),  
Comment TEXT,  
ReviewDate TEXT NOT NULL DEFAULT (datetime('now')),  
FOREIGN KEY (UserID) REFERENCES Users(UserID),  
FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)  
);
```

ActivityLogs Table:

```
CREATE TABLE IF NOT EXISTS ActivityLogs (  
LogID INTEGER PRIMARY KEY AUTOINCREMENT,  
UserID INTEGER NOT NULL,  
Action TEXT NOT NULL,  
Details TEXT,  
LogDate TEXT NOT NULL DEFAULT (datetime('now')),
```

```
IPAddress TEXT DEFAULT '127.0.0.1',  
FOREIGN KEY (UserID) REFERENCES Users(UserID)  
);
```

Seed Data Queries

Insert Default Roles:

```
INSERT INTO Roles VALUES (1,'SuperAdmin');  
INSERT INTO Roles VALUES (2,'Admin');  
INSERT INTO Roles VALUES (3,'User');
```

Insert Default Users:

-- Super Admin

```
INSERT INTO Users  
(Name,Email,PasswordHash,Phone,RoleID,LimitPerMonth,Status)  
VALUES ('Super  
Admin','admin@fuelms.com','[hashed_password]','01700000000',1,9999,'Active');
```

-- Sample Admin

```
INSERT INTO Users  
(Name,Email,PasswordHash,Phone,RoleID,LimitPerMonth,Status)  
VALUES ('John  
Admin','john@fuelms.com','[hashed_password]','01711111111',2,9999,'Active')  
;
```

-- Sample Customer

```
INSERT INTO Users
(Name,Email>PasswordHash,Phone,RoleID,LimitPerMonth>Status)
VALUES ('John
Doe','john@example.com','[hashed_password]','01722222222',3,100,'Active');
```

Insert Fuel Types:

```
INSERT INTO Fuel (FuelName,PricePerLiter,Quantity>Status) VALUES
('Petrol',110.00,2500,'Available');
```

```
INSERT INTO Fuel (FuelName,PricePerLiter,Quantity>Status) VALUES
('Diesel',100.00,2500,'Available');
```

```
INSERT INTO Fuel (FuelName,PricePerLiter,Quantity>Status) VALUES
('Octane',130.00,800,'Available');
```

Fuel Management Queries

Get All Fuel:

```
```sql
SELECT * FROM Fuel ORDER BY FuelName
```
```

3.3.2 Add New Fuel

```
```sql
INSERT INTO Fuel (FuelName,PricePerLiter,Quantity>Status)
VALUES (@n, @p, @q, @s)
```
```

Update Fuel:

UPDATE Fuel

SET FuelName=@n, PricePerLiter=@p, Quantity=@q, Status=@s

WHERE FuelID=@id

Delete Fuel (Soft Delete):

UPDATE Fuel SET Status='Unavailable' WHERE FuelID=@id

Update Stock (Add Quantity):

UPDATE Fuel SET Quantity = Quantity + @q WHERE FuelID=@id

Order Management Queries

Place Order:

-- Check fuel availability:

SELECT * FROM Fuel WHERE FuelID=@id AND Status='Available'

-- Get user's monthly limit:

SELECT LimitPerMonth FROM Users WHERE UserID=@id

-- Calculate used quantity this month:

SELECT COALESCE(SUM(o.Quantity),0) FROM Orders o

WHERE o.UserID=@uid AND o.Status != 'Cancelled'

AND strftime('%Y-%m', o.CreatedAt) = strftime('%Y-%m', 'now')

-- Insert order

```
INSERT INTO Orders (UserID,FuelID,Quantity,TotalPrice,Status)
VALUES (@u, @f, @q, @t, 'Pending')
```

-- Deduct from stock:

```
UPDATE Fuel SET Quantity = Quantity - @q WHERE FuelID=@id
...
```

Get Orders (with JOINS):

```
SELECT o.*, u.Name as UserName, f.FuelName
FROM Orders o
JOIN Users u ON o.UserID=u.UserID
JOIN Fuel f ON o.FuelID=f.FuelID
WHERE 1=1
AND o.UserID=@uid -- optional filter
AND o.Status=@st -- optional filter
ORDER BY o.CreatedAt DESC
...
```

Update Order Status:

```
,
UPDATE Orders SET Status=@s WHERE OrderID=@id
-- If cancelled, restore stock
SELECT FuelID, Quantity FROM Orders WHERE OrderID=@id
UPDATE Fuel SET Quantity=Quantity+@q WHERE FuelID=@fid
...
```

Get User Limit Info:

-- Get limit

```
SELECT LimitPerMonth FROM Users WHERE UserID=@id
```

-- Get used this month

```
SELECT COALESCE(SUM(Quantity),0) FROM Orders
```

```
WHERE UserID=@uid AND Status!='Cancelled'
```

```
AND strftime('%Y-%m', CreatedAt) = strftime('%Y-%m', 'now')
```

...

Payment Queries

Make Payment:

```
INSERT INTO Payments
```

```
(OrderID,UserID,Amount,PaymentMethod,Status,PaymentDate)
```

```
VALUES (@oid, @uid, @amt, @mth, 'Completed', datetime('now'))
```

Approve Order (with Payment):

```
UPDATE Orders SET Status='Approved' WHERE OrderID=@id AND  
Status='Pending'
```

Get Payments:

```
SELECT p.*, u.Name as UserName
```

```
FROM Payments p
```

```
JOIN Users u ON p.UserID=u.UserID
WHERE p.UserID=@uid -- optional filter
ORDER BY p.PaymentDate DESC
```

Check if Order has Payment

```
SELECT COUNT(*) FROM Payments WHERE OrderID=@oid
```

Review Queries:

Add Review:

Check for duplicate:

```
SELECT COUNT(*) FROM Reviews WHERE UserID=@u AND OrderID=@o
```

-- Insert review:

```
INSERT INTO Reviews (UserID,OrderID,Rating,Comment)
```

```
VALUES (@u, @o, @r, @c)
```

```
...
```

Get Reviews:

```
SELECT r.*, u.Name as UserName
```

```
FROM Reviews r
```

```
JOIN Users u ON r.UserID=u.UserID
```

```
WHERE r.UserID=@uid -- optional filter
```

ORDER BY r.ReviewDate DESC

Get Reviewed Order IDs:

```
SELECT OrderID FROM Reviews WHERE UserID=@uid
```

Activity Log Queries

Log Activity:

```
INSERT INTO ActivityLogs (UserID,Action,Details)
VALUES (@u, @a, @d)
```

Get Logs:

```
SELECT l.*, u.Name as UserName
FROM ActivityLogs l
JOIN Users u ON l.UserID=u.UserID
ORDER BY l.LogDate DESC
LIMIT 200
```

...

Dashboard Statistics Queries

Get Dashboard Stats:

```
SELECT
    (SELECT COUNT(*) FROM Users WHERE RoleID=3 AND Status='Active') as
    TotalUsers,
```


(SELECT COUNT(*) FROM Users WHERE RoleID IN(1,2) AND Status='Active')
as TotalAdmins,

(SELECT COUNT(*) FROM Orders) as TotalOrders,

(SELECT COALESCE(SUM(Amount),0) FROM Payments WHERE
Status='Completed') as TotalRevenue,

(SELECT COUNT(*) FROM Orders WHERE Status='Pending') as
PendingOrders,

(SELECT COALESCE(SUM(Amount),0) FROM Payments
WHERE Status='Completed' AND date(PaymentDate)=date('now')) as
TodaySales

Management Queries

User Login:

SELECT * FROM Users WHERE Email=@email AND Status='Active'

Register User:

INSERT INTO Users

(Name,Email,PasswordHash,Phone,RoleID,LimitPerMonth,Status)

VALUES (@n, @e, @p, @ph, @r, @l, 'Active')

...

Update User:

UPDATE Users

SET Name=@n, Email=@e, Phone=@ph, RoleID=@r, LimitPerMonth=@l,
Status=@s

WHERE UserID=@id

Change Password:

UPDATE Users SET PasswordHash=@p WHERE UserID=@id AND Email=@e

7. Transition Table

Login

Fuel Management System — Login

 **FuelMS**

FUEL MANAGEMENT SYSTEM

Sign In

EMAIL ADDRESS

PASSWORD

SIGN IN

[Forgot Password?](#)

Don't have an account?

Create Account

Forget Password

Reset Password

Reset Password

EMAIL ADDRESS

NEW PASSWORD

CONFIRM PASSWORD

UPDATE PASSWORD

Register New Account

Register New Account — □ ×

Create Account

FULL NAME

EMAIL ADDRESS

PHONE NUMBER

PASSWORD

CONFIRM PASSWORD

CREATE ACCOUNT

Customer Dashboard

FuelIMS
CUSTOMER

S.M. Tanvir Hassan
User

Dashboard

Fuel List

Place Order

Order History

Payments

My Limit

Reviews

Logout

Welcome, S.M. Tanvir Hassan!

Monthly Limit

Limit: 100L
Used: 30L

Remaining: 70L

Quick Actions

Place Order

Order History

My Recent Orders

| ID | Fuel | Qty | Total | Status |
|----|--------|-----|----------|----------|
| 3 | Diesel | 10L | ৳1000.00 | Approved |
| 2 | Diesel | 10L | ৳1000.00 | Approved |
| 1 | Diesel | 10L | ৳1000.00 | Approved |

Last Order

Fuel: Diesel
Qty: 10L
Amount: ৳1000.00
Status: Approved
Date: 01-05-2026

Super Admin Dashboard

FuelIMS

SUPER ADMIN

Super Admin

SuperAdmin

Dashboard

Manage Admins

Manage Users

Fuel Mgmt

Orders

Payments

Reports

System Settings

Activity Logs

Logout

Dashboard Overview

Refresh

TOTAL USERS

3

TOTAL ADMINS

2

TOTAL ORDERS

9

REVENUE (₹)

₹12,300

PENDING

0

Recent Orders

| ID | User | Fuel | Qty |
|----|-------|--------|-----|
| 9 | tvv | Diesel | 10L |
| 8 | tvv | Diesel | 10L |
| 7 | tvv | Octane | 10L |
| 6 | tavir | Diesel | 40L |
| 5 | tavir | Diesel | 10L |
| 4 | tavir | Diesel | 10L |
| 3 | SAIT | Diesel | 10L |

Fuel Stock

Diesel

2390L | ₹100/

Octane

1060L | ₹130/

Petrol

2500L | ₹110/

Admin Dashboard

FuelIMS

ADMIN

John Admin
Admin

Dashboard

Users

Fuel Mgmt

Orders

Payments

Reports

Reviews

Logout

Dashboard Overview

Refresh

TOTAL USERS

3

TOTAL ADMINS

2

TOTAL ORDERS

9

REVENUE (₹)

₹12,300

PENDING

0

Recent Orders

| ID | User | Fuel | Qty |
|----|-------|--------|-----|
| 9 | tvv | Diesel | 10L |
| 8 | tvv | Diesel | 10L |
| 7 | tvv | Octane | 10L |
| 6 | tavir | Diesel | 40L |
| 5 | tavir | Diesel | 10L |
| 4 | tavir | Diesel | 10L |

Fuel Stock

Diesel

2390L | ₹100/

Octane

1060L | ₹130/

Petrol

2500L | ₹110/

Customer Review

FuelMS

CUSTOMER

S.M. Tanvir Hassan

User

Dashboard

Fuel List

Place Order

Order History

Payments

My Limit

★ Reviews

Logout

Reviews

+ Add Review

↻ Refresh

| ID | User | OrderID | Rating | Comment | Date |
|----|------|---------|--------|---------|------|
|----|------|---------|--------|---------|------|

Add Review

Write a Review

SELECT ORDER

#3 — Diesel 10L — ₹1000.00

RATING (1-5)

5

COMMENT

SUBMIT REVIEW

Customer Order

FuelMS

CUSTOMER

S.M. Tanvir Hassan

User

Dashboard

Fuel List

Place Order

Order History

Payments

My Limit

Reviews

Logout

Place New Order

Order Details

Monthly Limit: 100L | Used: 30L | Remaining: 70L

SELECT FUEL TYPE

Diesel — ₹100/L (Stock: 2390L)

QUANTITY (Liters)

10

Total: ₹1000.00

PAYMENT METHOD

Cash

PLACE ORDER PAY

Fuel Price List

Diesel

₹100/L | 2390L

Octane

₹130/L | 1060L

Petrol

8.Conclusion

In conclusion, the Fuel Management System makes fuel-related processes easier and more efficient. Users can view fuel, place orders, make payments, and give reviews, while the admin manages fuel, users, orders, and payments. The superadmin has the full control over admin and user.

Overall, the system reduces manual work, saves time, and improves accuracy, making it a reliable and effective solution for fuel management.